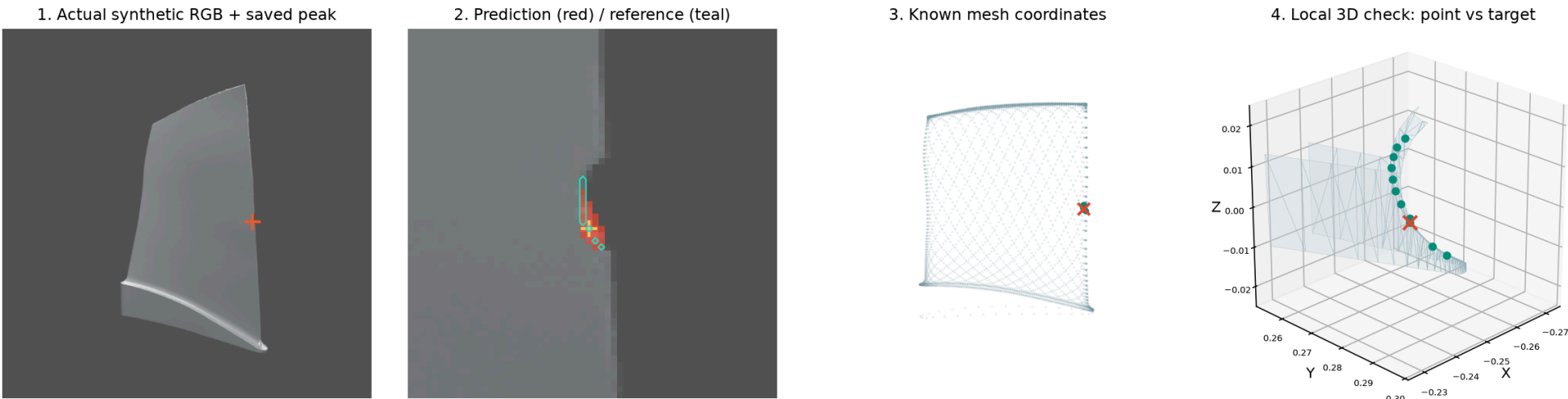


A correct 3D point, but an incomplete mask

The saved peak lands on the known exposed nick surface. Its mask IoU is only 20%: a correct point does not mean the entire damage area was segmented correctly.



Red/orange = saved model output | Teal = known synthetic target | Transparent mesh context is not an occlusion test.

The recorded result

Image alarm / score	Flagged / 11.657 (threshold 5.276)
Mask IoU / exact peak hit	20.00% / yes
3D reference	remaining damaged mesh
Distance to sampled target	0.00000000 normalized units

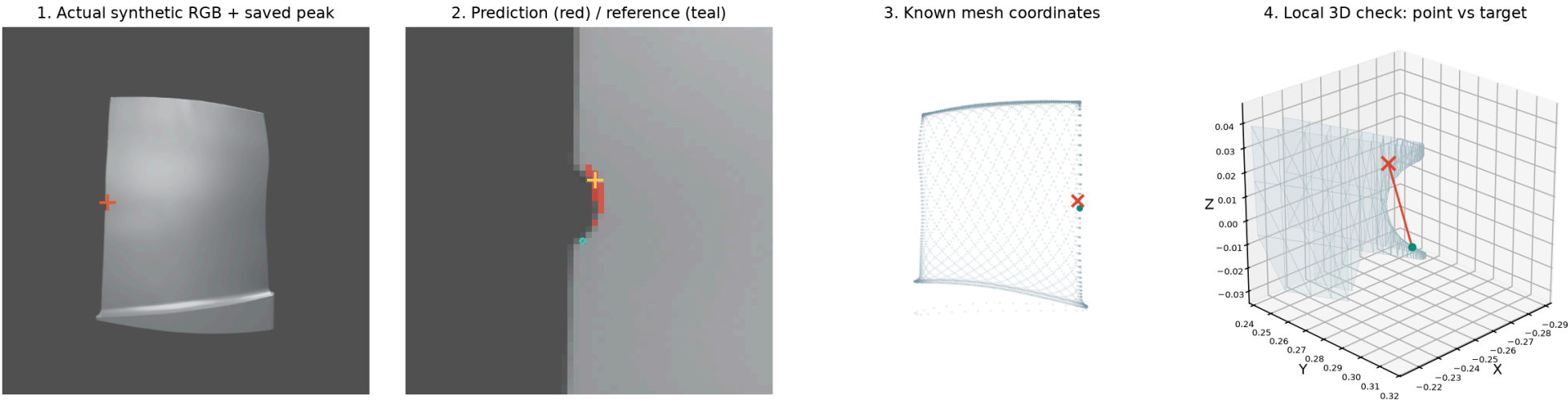
The check you can inspect

Saved peak, zero-based pixel	[347, 267]
3D coordinate (X, Y, Z)	-0.250668422, 0.275741425, -0.004085742
Reference triangle	60247
Reprojection residual	0.00000001 pixels

What this proves: the displayed outcome can be reproduced from actual saved predictions and known synthetic mesh/camera records. **What it does not prove:** a detector inferred the mesh, generalizes to other blades, or achieves verified real-world 3D accuracy.

A valid surface point can still be wrong

The point can be reconstructed on the remaining blade, but it misses the labeled damage. Geometric mapping succeeds while damage localization fails.



Red/orange = saved model output | Teal = known synthetic target | Transparent mesh context is not an occlusion test.

The recorded result

Image alarm / score	Flagged / 8.271 (threshold 5.276)
Mask IoU / exact peak hit	0.00% / no
3D reference	remaining damaged mesh
Distance to sampled target	0.03467299 normalized units

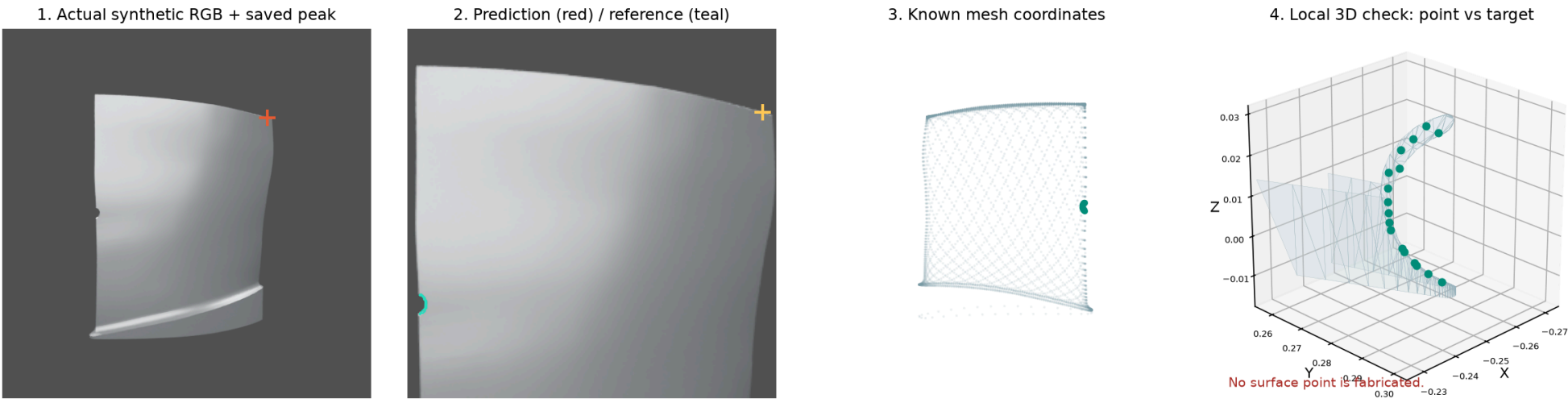
The check you can inspect

Saved peak, zero-based pixel	[147, 245]
3D coordinate (X, Y, Z)	-0.249747163, 0.274580646, 0.023097913
Reference triangle	39978
Reprojection residual	0.00000154 pixels

What this proves: the displayed outcome can be reproduced from actual saved predictions and known synthetic mesh/camera records. **What it does not prove:** a detector inferred the mesh, generalizes to other blades, or achieves verified real-world 3D accuracy.

Background must stay unmapped

The saved peak falls on background. There is no corresponding surface triangle, so the output remains unmapped. The image alarm is also negative; a raw mask peak is not itself an alarm.



Red/orange = saved model output | Teal = known synthetic target | Transparent mesh context is not an occlusion test.

The recorded result

Image alarm / score	No alarm / -2.129 (threshold 5.276)
Mask IoU / exact peak hit	0.00% / no
3D reference	remaining damaged mesh
Distance to sampled target	Undefined

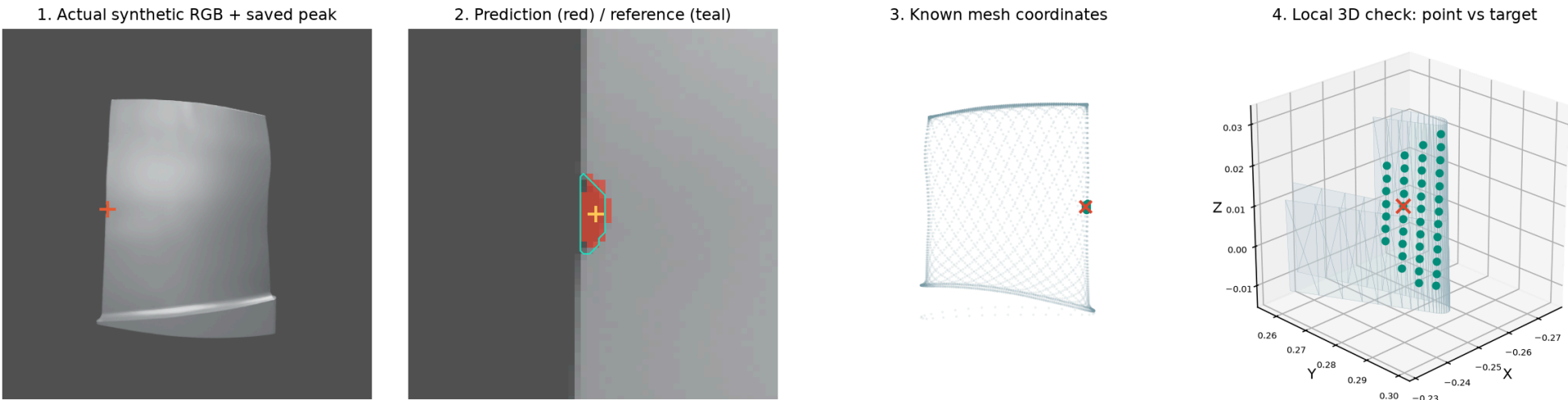
The check you can inspect

Saved peak, zero-based pixel	[368, 123]
3D coordinate (X, Y, Z)	Unavailable — source pixel has no surface mapping
Reference triangle	None
Reprojection residual	Not applicable

What this proves: the displayed outcome can be reproduced from actual saved predictions and known synthetic mesh/camera records. **What it does not prove:** a detector inferred the mesh, generalizes to other blades, or achieves verified real-world 3D accuracy.

Missing material needs the intact reference

This A-head peak is checked against where the intact surface used to be. It is a reference association for missing material, not an invented point on the remaining nick surface.



Red/orange = saved model output | Teal = known synthetic target | Transparent mesh context is not an occlusion test.

The recorded result

Image alarm / score	Flagged / 8.271 (threshold 5.276)
Mask IoU / exact peak hit	70.59% / yes
3D reference	intact reference
Distance to sampled target	0.00000000 normalized units

The check you can inspect

Saved peak, zero-based pixel	[146, 249]
3D coordinate (X, Y, Z)	-0.252416148, 0.277117538, 0.009930615
Reference triangle	40114
Reprojection residual	0.00000068 pixels

What this proves: the displayed outcome can be reproduced from actual saved predictions and known synthetic mesh/camera records. **What it does not prove:** a detector inferred the mesh, generalizes to other blades, or achieves verified real-world 3D accuracy.

We have working examples. We do not yet have proof of the full research goal.

Implemented and checked

- Actual public synthetic blade RGB, saved neural scores and native binary masks.
- Separate references for missing material (A) and exposed remaining surface (B).
- Image pixel → triangle → barycentric coordinate → known 3D point.
- Reprojection and target-membership checks; wrong and unmapped points stay visible.

```
point = w0 × vertex0 + w1 × vertex1 + w2 × vertex2
```

The JSON includes the three vertices, weights, point coordinates, selected pixels and source hashes. These examples replay existing work; no new neural inference, training or geometry generation was performed.

The four displayed examples are selected to explain distinct outcomes from one previously examined public synthetic blade, one saved model seed and original thresholds. They are not a new benchmark, a random sample, a physical measurement or a claim that 3D geometry was inferred correctly from RGB. Reference labels determine the explanatory zoom crops; these crops are not model inputs.

Reproducibility and source

[evidence.json](#) records verified source hashes and full-precision calculations. Rebuild from the repository using `PYTHONPATH=src:scripts ../venv_linux/bin/python scripts/build_main_jet_visual_evidence.py --pdf`.

Source geometry: [Safran / PLAID Rotor37](#), sample 0, revision bac06c0caa7254120eccc6711a5fb85c58dfdbdc. Existing project alterations: normalization, artificial collar, sphere-cut nick and synthetic rendering. Our figures add saved predictions, reference comparisons and transparent mesh displays. Derived figures: [CC BY-SA 4.0](#); see [attribution](#). No confidential industrial photographs or private reconstruction assets are embedded.

The earlier full-geometry check failed

The separate 20 September direct-tracking study supplied source queries and exact synthetic cameras. It was not an automatic damage-detection evaluation.

Primary-pair population	Within 0.01 normalized units	Outside
Nick queries	78 / 82	4 / 82
Other surface queries	131 / 198	67 / 198

The complete case failed its surface criterion. Two cases were initially blocked by memory. They completed on 21 September: all three full-surface checks failed. A separate detector-selected-point integration now has three successful examples; [see the subsequent work](#). Neighboring queries on one nick are not independent defects.

[Recorded tracking findings](#) · [Real-photo geometry limitations](#)